Math 152 Problem Set 5

Due 11/6/2026, at 10:00 PM

Problem 1. Let q be a positive rational number. Consider the region below the graph of x^q over the interval $[0, R]$. This region can be revolved around the x-axis or the y-axis to obtain two solids.

(a) Suppose that $q < 1$. Which of the two solids will be larger when R is very large?[1]

(b) Which will be larger if $q > 1$, and R is very large?

(c) What happens when $q = 1$?

Problem 2. Consider a cone with base radius r and height h. What is the average area $A(x)$ of the vertical cross section at x of the cone, as x ranges over $[-r, r]$?

Problem 3. Recall that for a function f on $[a, b]$, we use $\overline{f}$ to denote its average value.

(a) Argue that for functions f and g on $[a, b]$, we have $\overline{f + g} = \overline{f} + \overline{g}$.

(b) Show by example that the average value of f^2, $\overline{f^2}$, may not be equal to $\left(\overline{f}\right)^2$.[2]

Problem 4. Give an example of a polynomial function $\mathbb{R} \to \mathbb{R}$ of degree greater than 1 which is invertible, and an example of a polynomial function which is not invertible. For the invertible function, find an expression for its inverse. For the function which is not invertible, explain why it is not invertible.

Problem 5. Consider the following accumulation function:

$$F(x) = \int_0^x \frac{2t^2 + 1}{\sqrt{t^4 + 1}} \mathrm{d}t$$

We view this as a function $F : \mathbb{R} \to \mathbb{R}$. By the fundamental theorem of calculus, it is continuous and differentiable.

(a) Argue that F is an increasing function, and therefore injective.

(b) Use the change of variable formula to show that F is an odd function.

(c) Argue[3] that for $x > 0$, $F(x) \geq x$ and for $x < 0$, $F(x) \leq x$. Conclude that F is unbounded in both directions, and therefore surjective.

(d) From (a) and (c), we know that F has an inverse $G : \mathbb{R} \to \mathbb{R}$. Argue that G satisfies the equation[4]

$$G'(x) = \frac{\sqrt{G(x)^4 + 1}}{2G(x)^2 + 1}$$

for all x.

[1] Hint: Compute the volume of each as a function of R, and compare them.

[2] It is always true that $\left(\overline{f}\right)^2 \leq \overline{f^2}$. This is an instance of what is called the Cauchy-Schwarz inequality.

[3] Hint: Argue that the function inside the integral is always ≥ 1.

[4] An equation like this relating the derivative of a function to values of the function is called a *differential equation*. We will study equations like this near the end of the course.

Problem 6 (Optional)**.** On the previous problem set, an optional problem asks you to find the volume of a torus obtained by revolving a disk of radius r centered at $(R, 0)$ around the y-axis. The volume comes out to $2\pi R \cdot \pi r^2$, which can be interpreted as the product of the area of the disk and the circumference of a circle of radius R. This problem will guide you through the derivation of a weak form of Pappus' Theorem[5], which can be used to derive many other similar formulas for volume.

(a) Suppose that a function $h(x)$ is symmetric across the line $x = R$, so $h(R - x) = h(R + x)$ for all x. Argue[6] that

$$\int_{R-a}^{R+a} x \cdot h(x)\mathrm{d}x = R \cdot \int_{R-a}^{R+a} h(x)\mathrm{d}x$$

(b) Suppose that a region with area A and a vertical line of symmetry $x = R$ is revolved around the y-axis. Argue that the volume of the resulting solid is $2\pi RA$.

[5]Pappus' Theorem states that the volume of a solid of revolution is the area of the revolved region multiplied by the circumference of the circle described by the *centroid* of the region. The centroid of a region is its center of mass, the point on which it would balance. For a symmetric region, this will always lie on the line of symmetry. See Chapter 6.4 of the book for centroids and Pappus' Theorem.

[6]Hint: Subtract the two integrals and combine them, use the substitution $u = x - R$, then apply symmetry.